

# Agile Requirement Gathering

Shola Oyedeji

# What is Agile

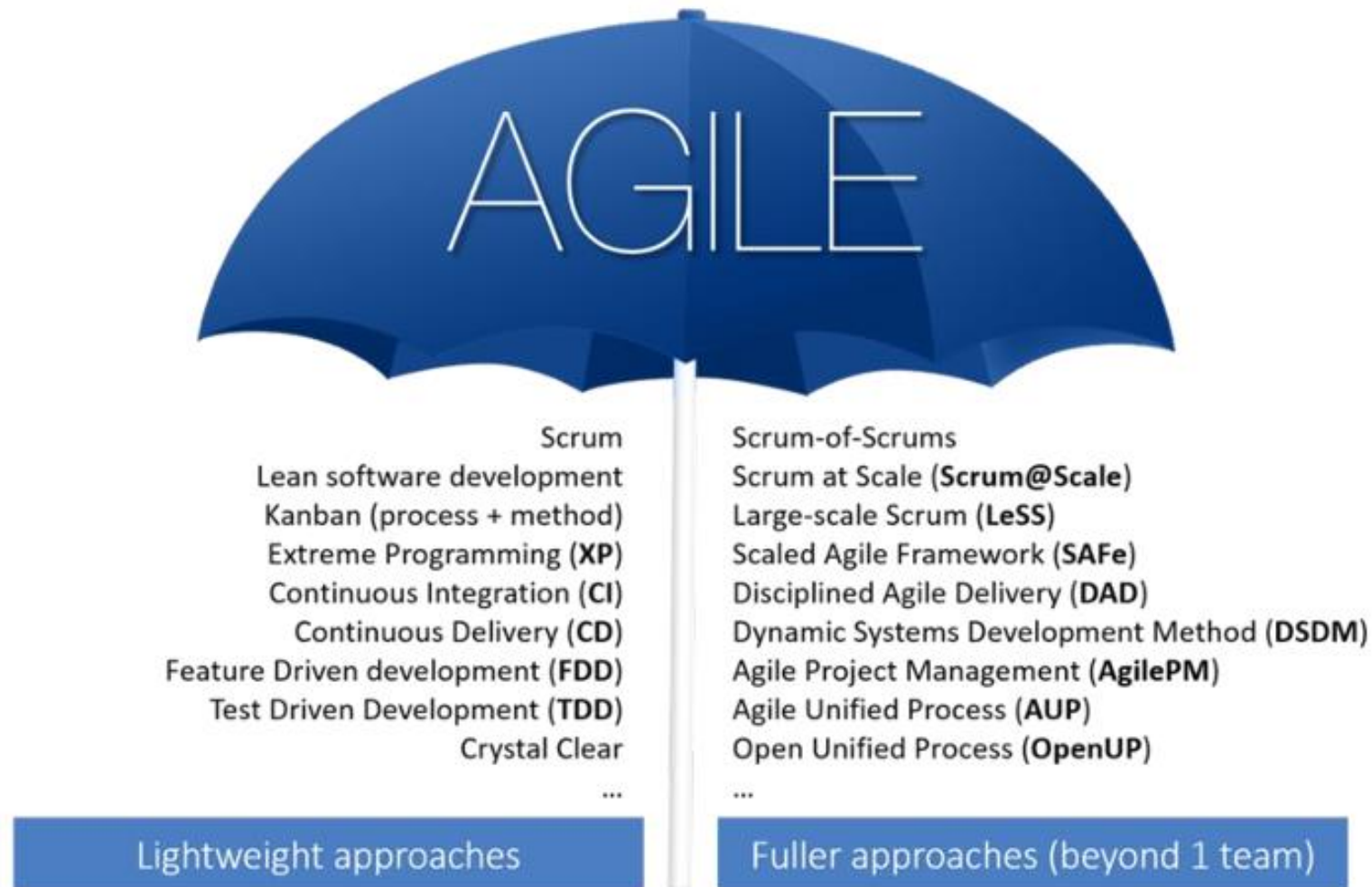
## AGILE IS...

- A group of development methodologies that are based on similar principles:
  - Alignment with customer needs and company goals
  - Frequent and fast delivery of high quality working product
  - Team work, self-organization and accountability
  - Frequent inspection and adaptation
- **Culture and organizational way of working**

## AGILE IS NOT...

- Undisciplined way of working
- Process
- Set of things
- A set of tools

# Agile Umbrella



# 12 Agile Principles

Satisfy the  
customer

Welcome  
Change

Deliver  
frequently

Collaborate  
daily

Support and  
trust motivated  
individuals &  
teams

Enable  
face-to-face  
conversation

Deliver  
working  
software

Promote  
sustainable  
pace

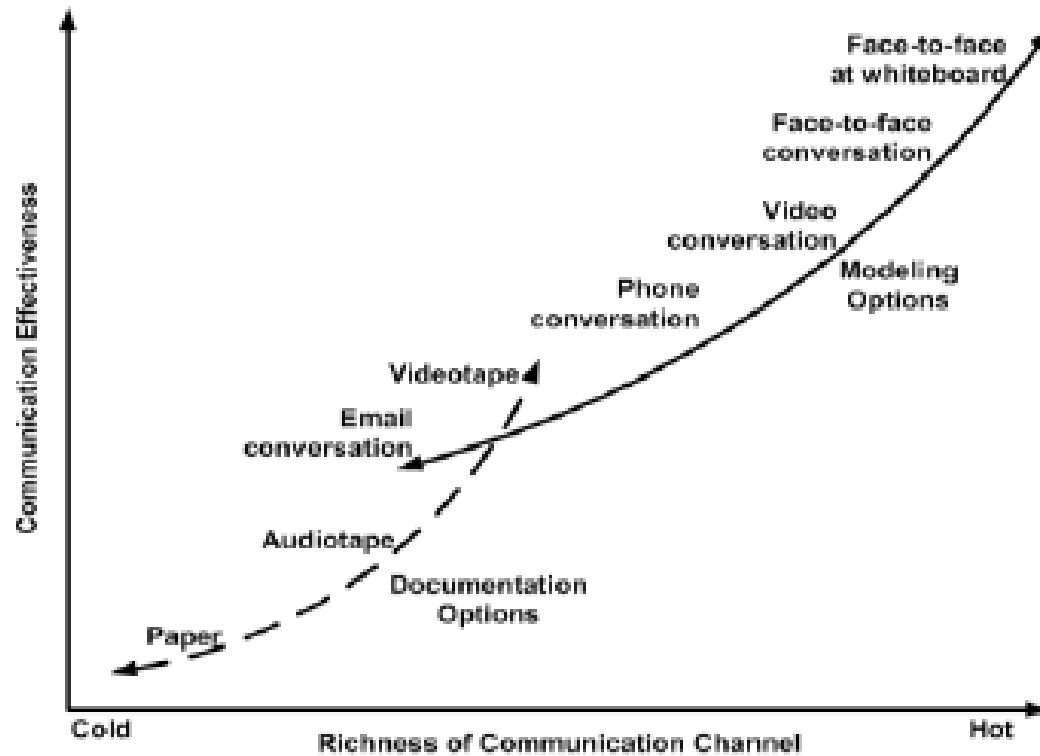
Promote  
technical  
excellence &  
good design

Maximize  
simplicity

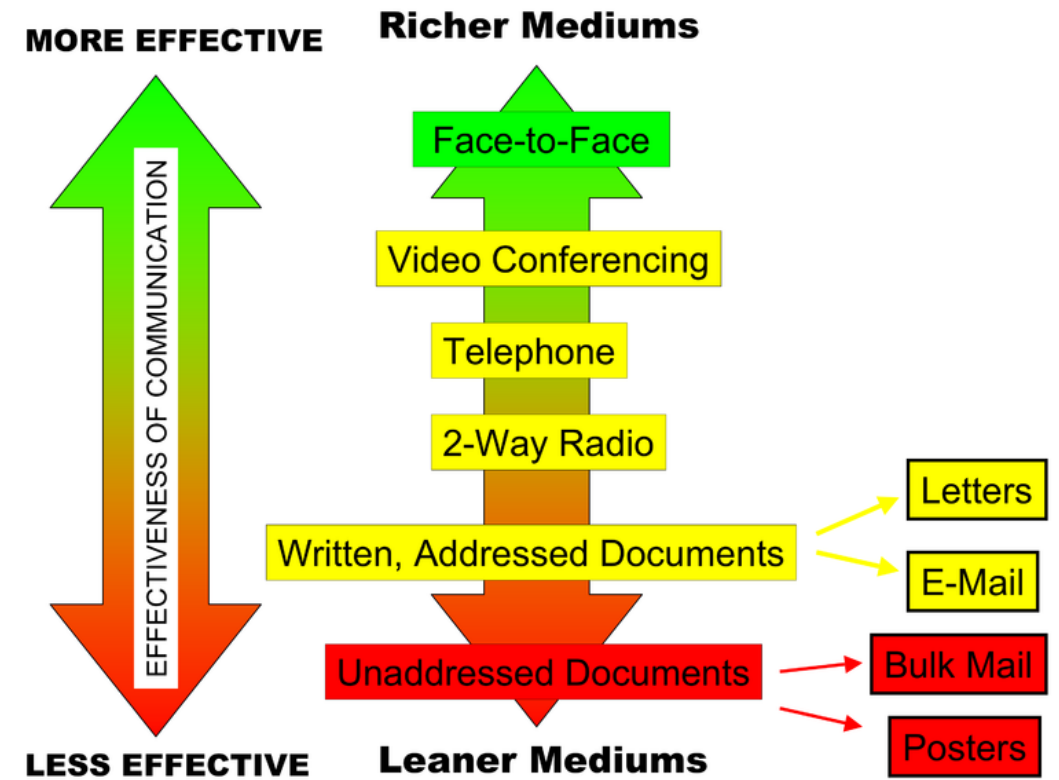
Grow  
self-organizing  
teams

Reflect and  
adjust regularly

# Communication



Copyright 2002-2005 Scott W. Ambler  
Original Diagram Copyright 2002 Alastair Cookburn



# Agile Requirement

## Conversion

- Conversation as a primary form of communication
- Face to face discussion between development team and customers, business owners, Vs Hand-off Requirement document



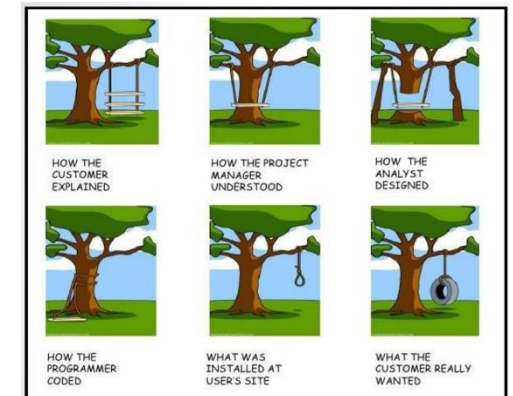
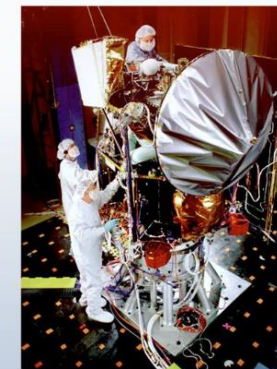
## Adaptive

- Discover user need Vs Collect User Needs

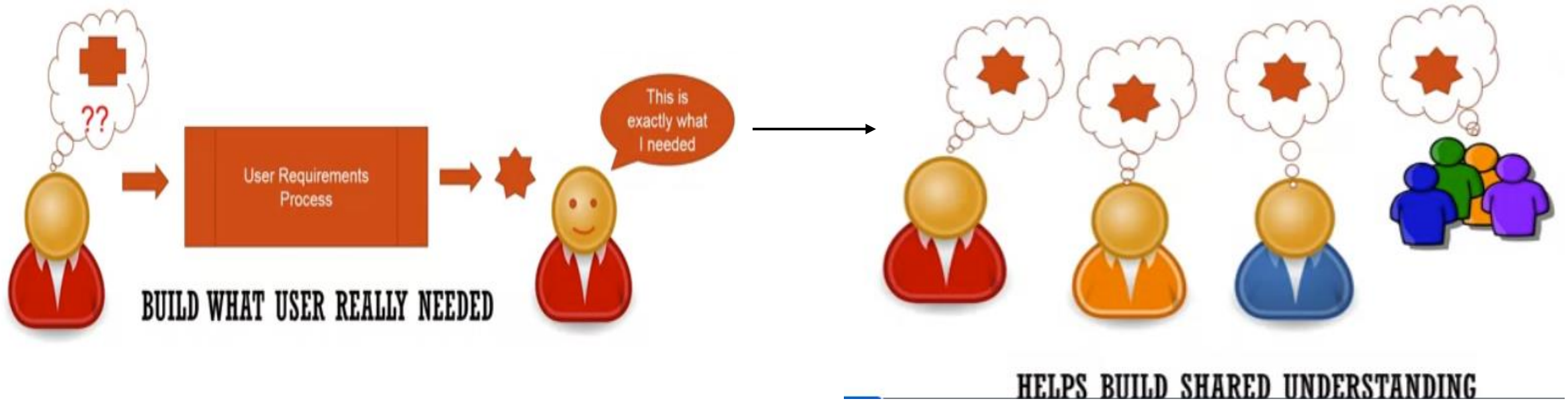


### The Mars Climate Orbiter (MCO)

In one of the all time great engineering mistakes, NASA's partner, Lockheed Martin, used English Standard units, not the metric units used by NASA. The probe eventually hit the Martian atmosphere at the wrong angle and burned up.



# User Requirement



# Agile Requirement



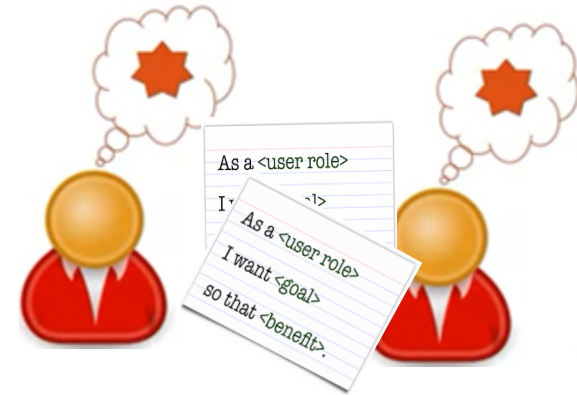
**CARD**



**CONVERSATION**



**CONFIRMATION**





# Agile Requirement



# User Story

A user story is an **end goal** not a **feature** written from the user or customer perspective. It is requirement for any functionality or feature which is written down in one or two lines.

As a <user role>

I want <goal>

so that <benefit>.

# User Story



Example of user story:

- Structure
- Acceptance Criteria

## Task:

- Write a user story for a user who wants to search to find matching hotel option in a hotel web page
- Acceptance Criteria

# User Story



**As a** <role>  
**I want** <goal>  
**so that** <benefit>

**Acceptance criteria:**  
(Conditions of Satisfaction)

...

...

**As an** Account Manager  
**I want** a sales report of my account to be sent to my inbox daily  
**So that** I can monitor the sales progress of my customer portfolio

## **Acceptance criteria:**

1. The report is sent daily to my inbox
  2. The report contains the following sales details: ...
  3. The report is in csv format.
- 

# User Story

A good user story must follow the **INVEST** principle:

- **Independent:** Each user story should stand on its own, independent from other stories.
- **Negotiable:** Stories are not contracts, rather opportunities for negotiation and change.
- **Valuable:** Every story should add value for users and stakeholders.
- **Estimable:** Every story's time and budget costs should be calculable, based on domain and technical knowledge.
- **Small:** User stories should be small enough to estimate and implement simply.
- **Testable:** Make sure you can test the user story through criteria the story itself explains.



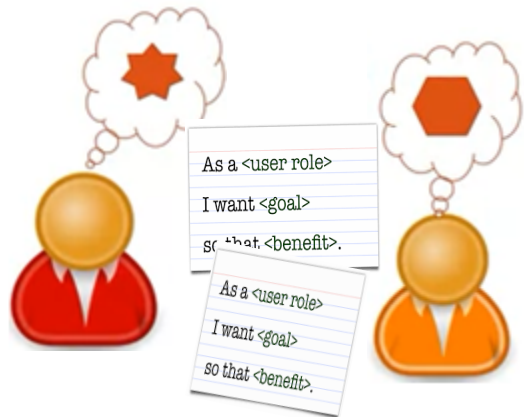
Proprietary



# Agile Requirement



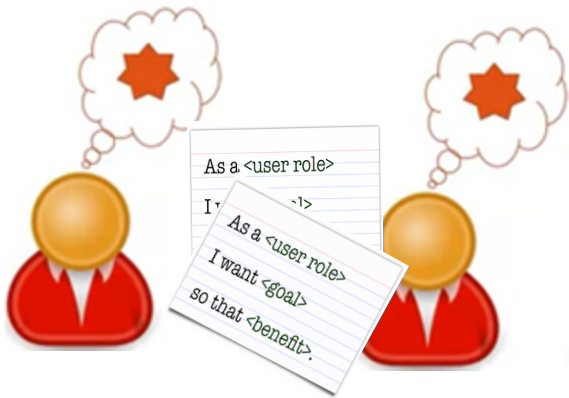
## CONVERSATION



# Agile Requirement



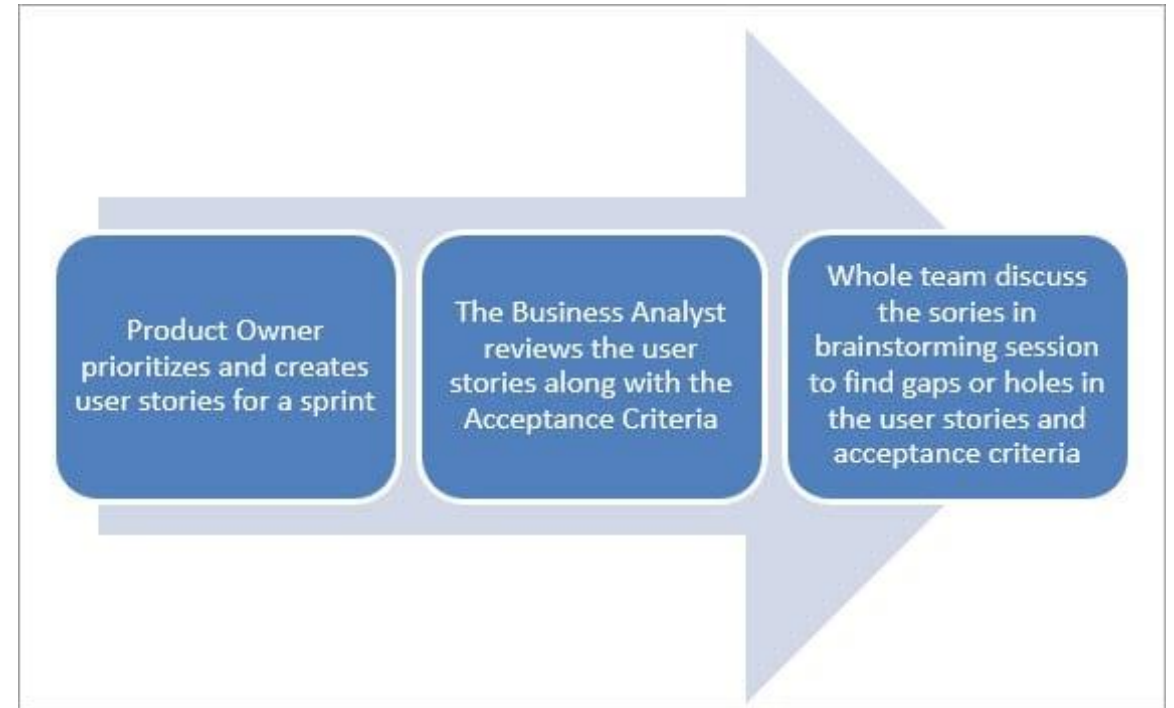
## CONFIRMATION



# Acceptance criteria (AC)

---

- Acceptance criteria (AC) are sets of accepted conditions or business rules which the functionality or feature should satisfy and meet in order to be accepted by product owner or stakeholder.

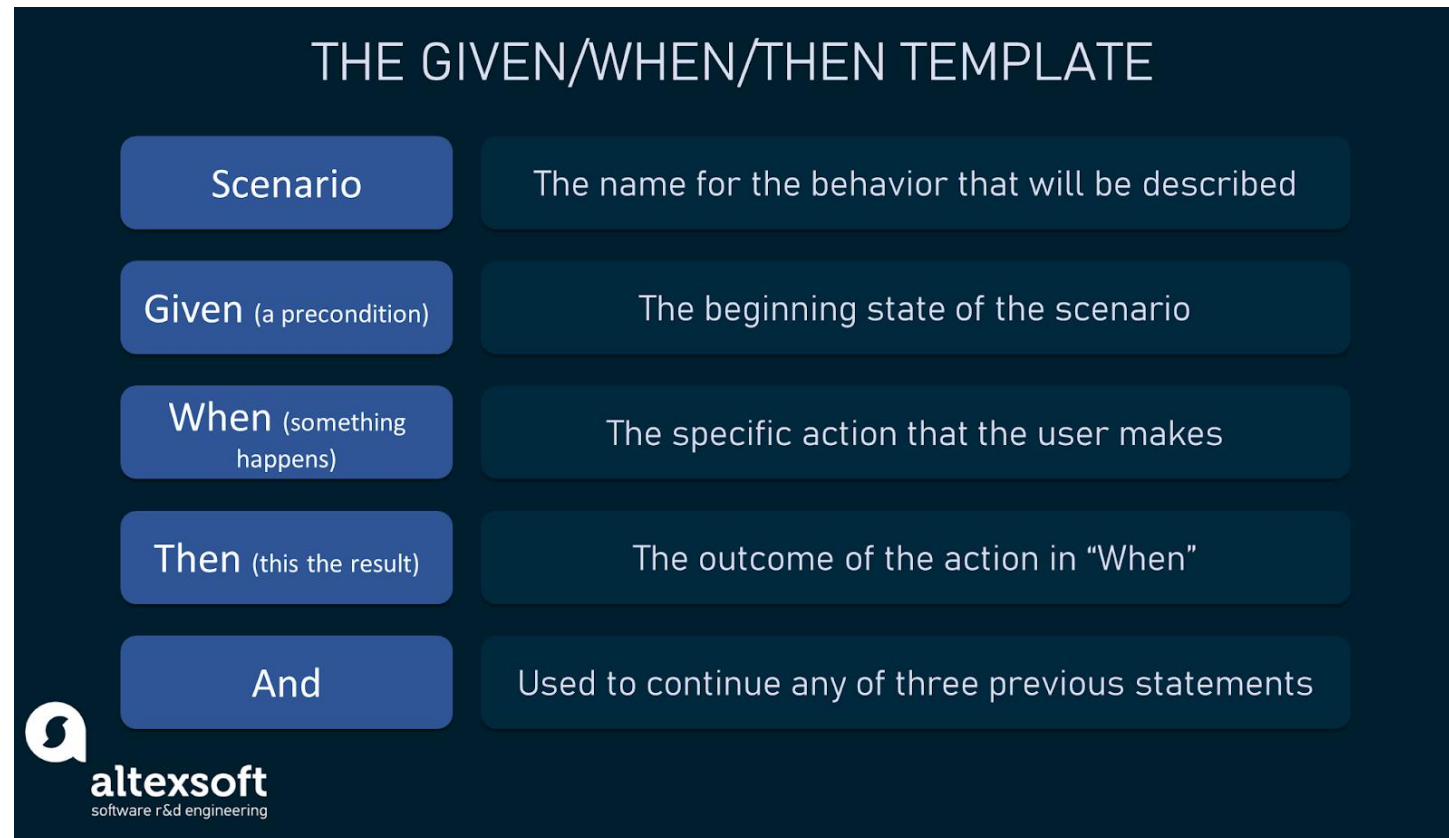


# Acceptance criteria (AC)

---

The acceptance criteria template includes:

- **Scenario** – the name for the behavior that will be described
- **Given** – the beginning state of the scenario
- **When** – specific action that the user makes
- **Then** – the outcome of the action in “When”
- **And** – used to continue any of three previous statements





# Acceptance criteria (AC) Examples

## THE GIVEN/WHEN/THEN ACCEPTANCE CRITERIA: EXAMPLE 2

User story: As a user, I want to be able to request the cash from my account at an ATM so that I will be able to receive the money from my account quickly and in different places.

### Scenario 1

Requesting the cash from a creditworthy account

Given

The account is creditworthy

And

The card is valid

And

The dispenser contains cash

When

The customer requests the cash

Then

Ensure the account is debited

And

Ensure cash is dispensed

And

Ensure the card is returned



altexsoft  
software r&d engineering

## THE GIVEN/WHEN/THEN ACCEPTANCE CRITERIA: EXAMPLE 2

User story: As a user, I want to be able to request the cash from my account at an ATM so that I will be able to receive the money from my account quickly and in different places.

### Scenario 2

Requesting the cash from an overdrawn account

Given

The account is overdrawn

And

The card is valid

When

The customer requests the cash

Then

Ensure the rejection message is displayed

And

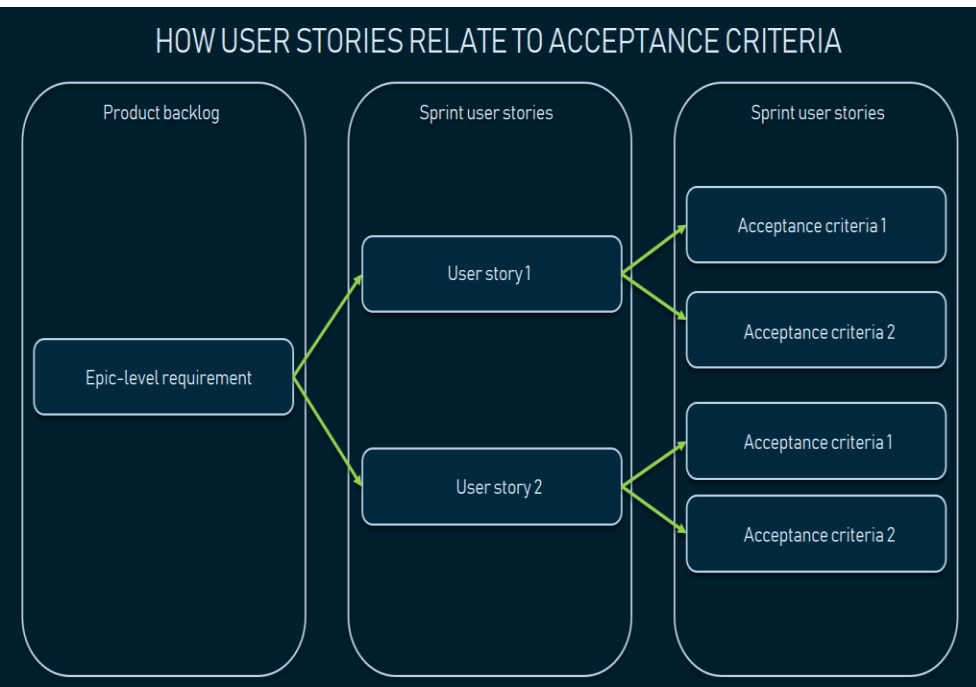
Ensure cash isn't dispensed



altexsoft  
software r&d engineering

# User Story and Acceptance Criteria

## HOW USER STORIES RELATE TO ACCEPTANCE CRITERIA



## User Story for Creating a New Account

### User Story

As a **new fintech app user**, I want to **create a new bank account** using the fintech mobile app, so that I can **automate my financial transactions**.

### Acceptance Criteria

- The user can open the app.
- The user is prompted a list of list of personal information questions.
- The user can submit further requirements for validation, such as government-issued IDs.
- The user receives a confirmation via email or SMS of their application.
- User can start using the app's banking features and start transacting.

# Task: Acceptance Criteria

## Task:

1. Write a user story for a user who wants to search to find matching hotel option in a hotel web page
2. Acceptance Criteria

## Result:

**User story:** *As a user, I want to use a search field to type a city, name, or street, so that I could find matching hotel options.*

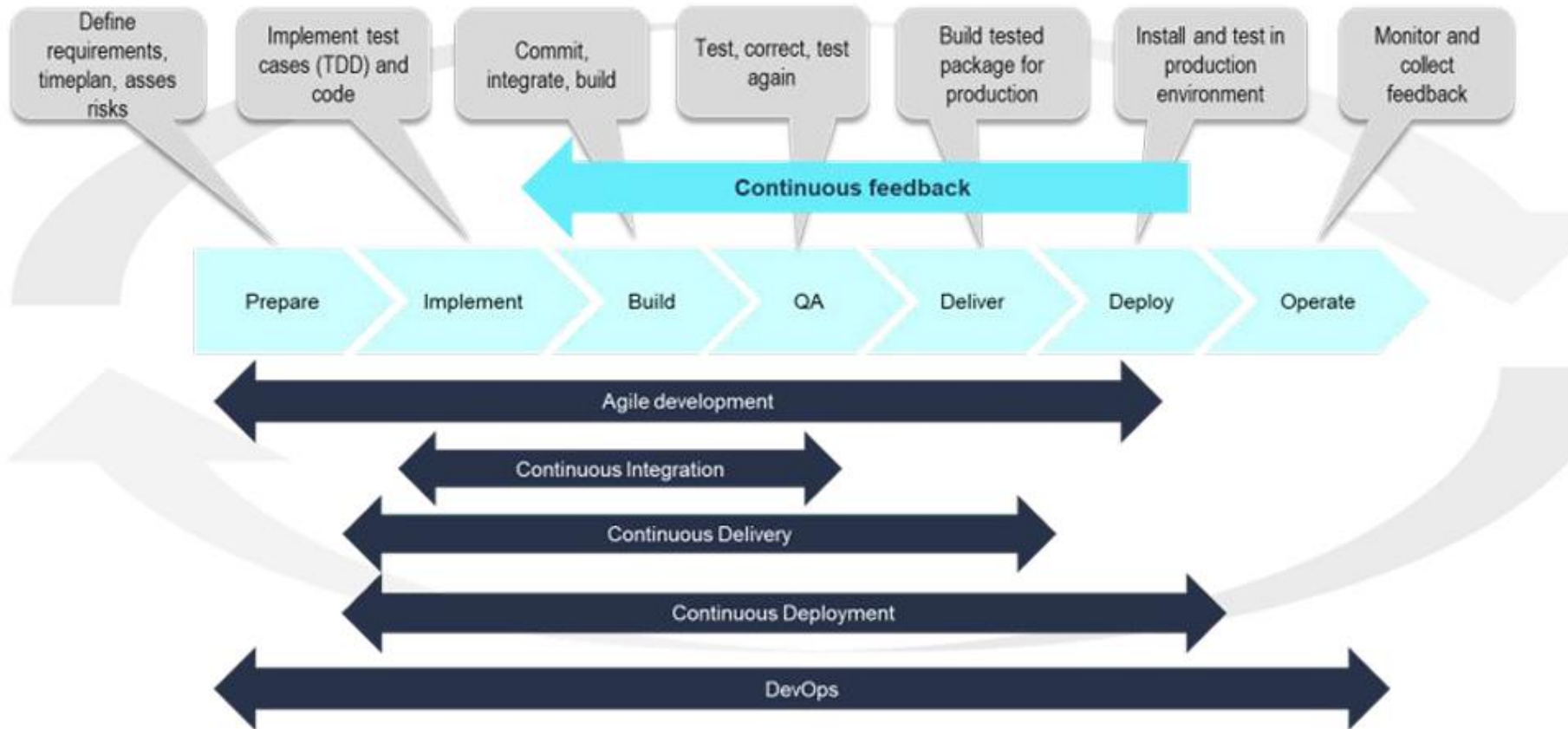
### **Basic search interface acceptance criteria**

- The search field is placed on the top bar
- Search starts once the user clicks “Search”
- The field contains a placeholder with a grey-colored text: “Where are you going?”
- The placeholder disappears once the user starts typing
- Search is performed if a user types in a city, hotel name, street, or all combined
- Search is in English, French, German, and Finnish
- The user can’t type more than 200 symbols

# Agile Development



# Agile Development

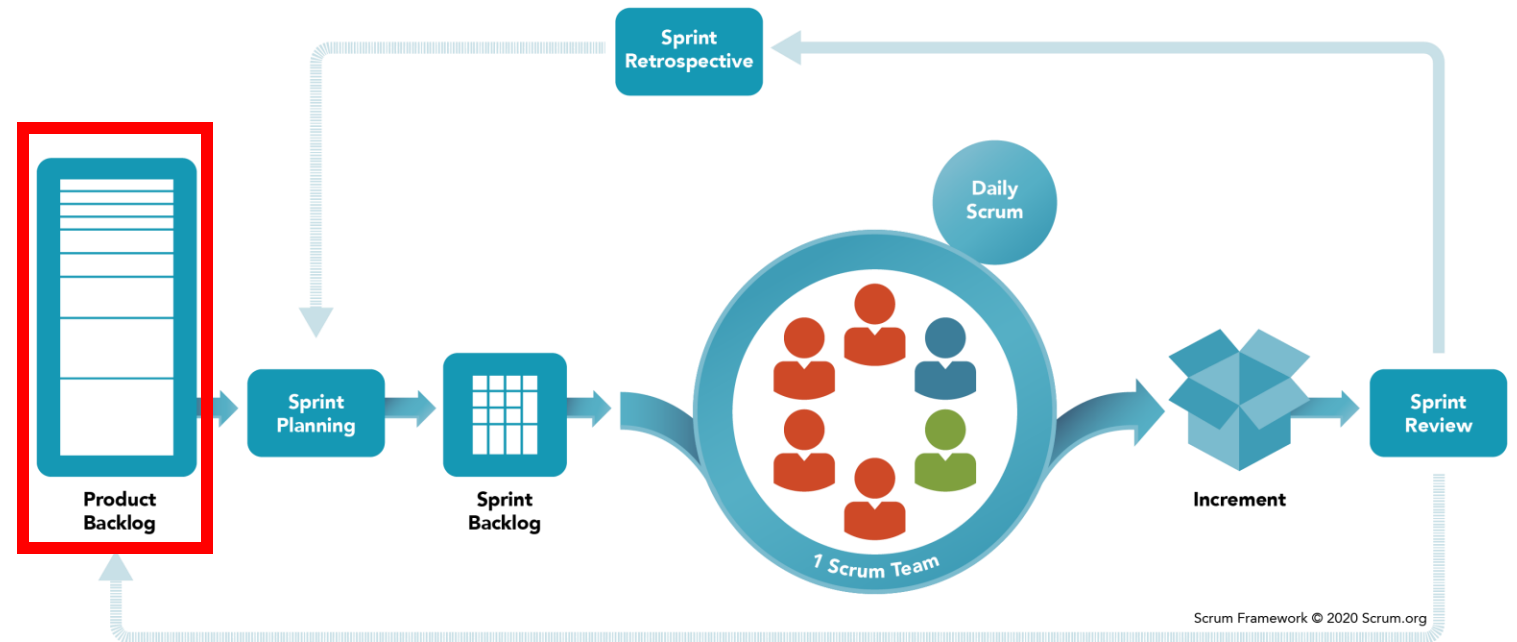


# What is Scrum?

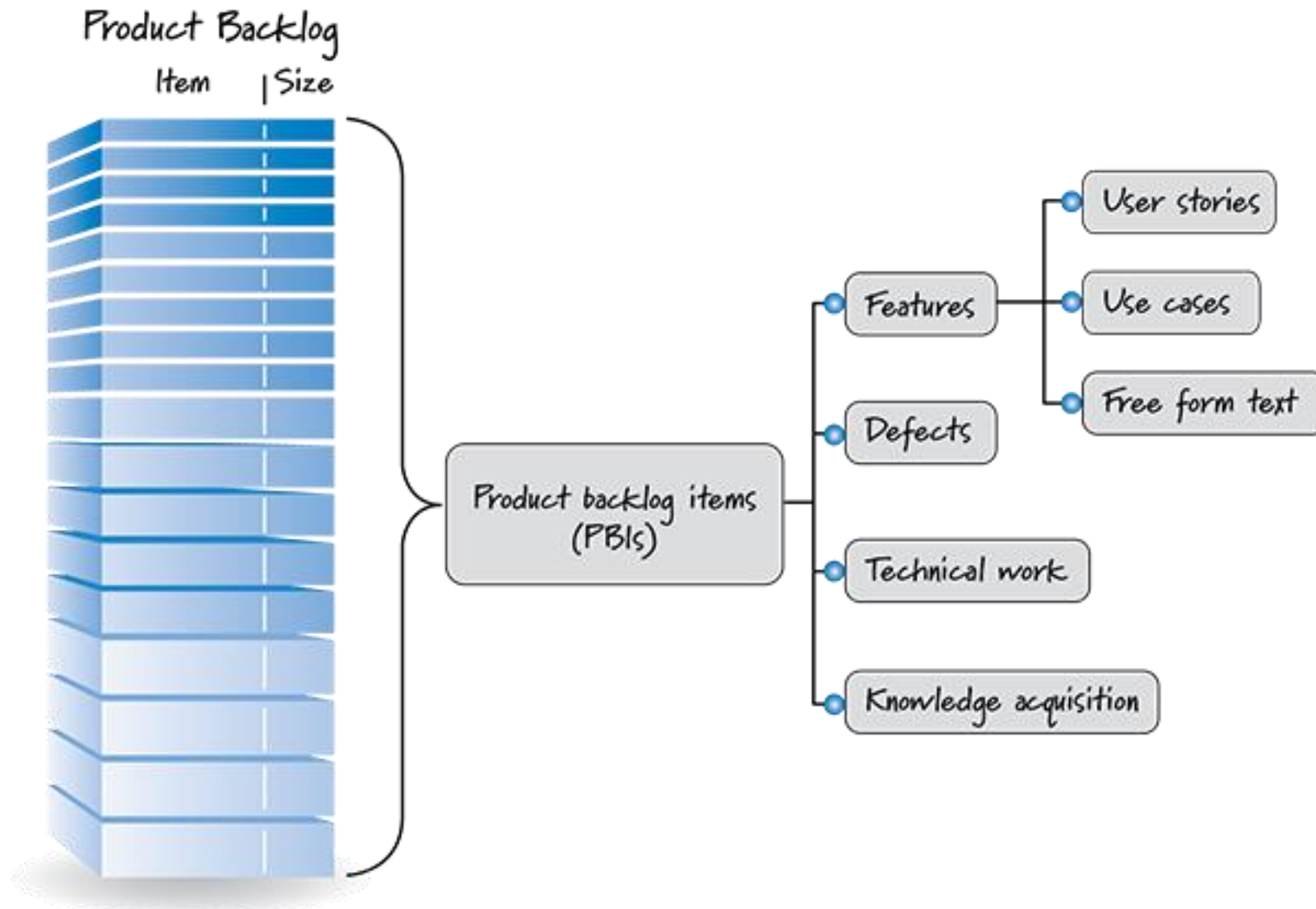


# Scrum

- Scrum is one of the most famous agile methodologies characterized by cycles or stages of development called sprints.
- Scrum is a framework used by people to address complex adaptive problems



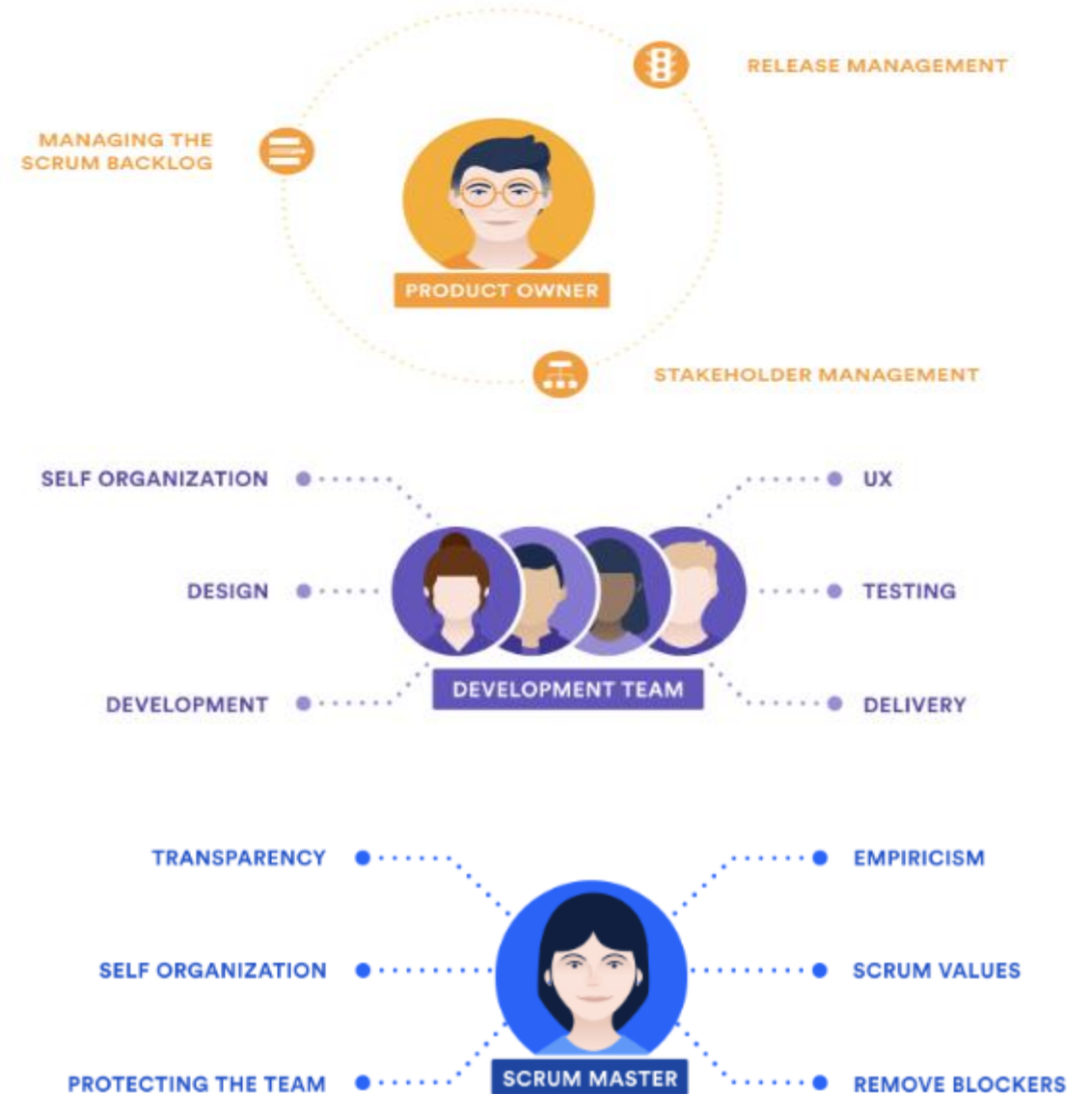
# Scrum: Product Backlog



# Scrum Roles, Artifacts and Events

| ROLES                                                                                                                                                                                                                                                                                                                                                                        | ARTIFACTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | EVENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Scrum Team</b><br/>Consists of one Scrum Master, one Product Owner, and Developers</p> <p><b>Product Owner</b><br/>Set Priorities<br/>Manage Product Backlog</p> <p><b>Scrum Master</b><br/>Teach Scrum<br/>Manage Process<br/>Protect Team<br/>Enforce Rules<br/>Remove Blocks</p> <p><b>Developers</b><br/>Develop Product<br/>Organize Work<br/>Update Progress</p> | <p><b>Product Backlog</b><br/>List of requirements<br/>Owned by the Product Owner<br/>Ordered by business value, dependency, risk<br/>Can change without affecting the active Sprint<br/>Commitment: Product Goal</p> <p><b>Sprint Backlog</b><br/>Decomposed task list<br/>Driven by a portion of Product Backlog<br/>Planned by and for the Developers<br/>Commitment: Sprint Goal</p> <p><b>Increment</b><br/>Version of the Product<br/>Potentially shippable<br/>Working functionality<br/>Commitment: DoD</p> | <p><b>Sprint</b><br/>Time-boxed to one month or less<br/>Potentially releasable product Increment is created</p> <p><b>Sprint Planning</b><br/>Time-boxed to max 8 hours for 1 month Sprint<br/>Team select what and how will be DONE in Sprint<br/>Sprint Goal</p> <p><b>Daily Scrum</b><br/>Time boxed 15 min / same time / place every day<br/>Check progress against Sprint Goal</p> <p><b>Sprint Review</b><br/>Time-boxed max 4 hours for 1 month Sprint<br/>Informal meeting with team and stakeholders<br/>Team present what is DONE and what is not</p> <p><b>Retrospective</b><br/>Time-boxed to max 3 hour for 1 month Sprint<br/>Team inspects itself and create a plan for improvements</p> |

# Scrum Roles



# Scrum Product Backlog - MVP Product Backlog for Netflix

---

| Product Backlog Item (PBI) | Description                                                                           | Priority |
|----------------------------|---------------------------------------------------------------------------------------|----------|
| User Registration & Login  | Enable users to securely create accounts and log in.                                  | High     |
| User Profiles              | Allow multiple profiles within a single account, supporting personalized experiences. | High     |
| Search & Browse Content    | Provide content search and browsing by category; include recommendations.             | High     |
| Play Video Content         | Support streaming of video content across devices.                                    | High     |
| Content Details Page       | Show detailed descriptions, trailers, and related content for each title.             | Medium   |
| Watchlist                  | Enable users to create a personalized watchlist.                                      | Medium   |
| Playback Controls          | Implement basic playback controls (pause, play, rewind, etc.).                        | Medium   |
| Basic Recommendations      | Offer content recommendations based on user viewing history.                          | Medium   |
| Content Licensing & DRM    | Ensure DRM protections are in place for licensed content.                             | High     |

# Sprint Backlog - Netflix

| <b>Sprint Backlog Item</b>    | <b>Description</b>                                               | <b>Estimated Time</b> |
|-------------------------------|------------------------------------------------------------------|-----------------------|
| <b>User Registration UI</b>   | Develop the front-end forms for user registration and login.     | 2 days                |
| <b>Backend Authentication</b> | Set up secure backend authentication for login and registration. | 3 days                |
| <b>User Profile Setup</b>     | Implement functionality for profile creation and management.     | 3 days                |
| <b>Content Search</b>         | Build the search functionality with basic filters.               | 4 days                |
| <b>Basic Video Playback</b>   | Develop video playback capabilities with essential controls.     | 5 days                |



How can you redesign scrum  
product backlog with  
consideration of sustainability ?

